Parent Function Families **TEACHER KEY**

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Parent function: the simplest form of a family — no shifts, stretches, or reflections.

Domain / Range: domain is the set of allowed x-values; range is the set of resulting y-values.

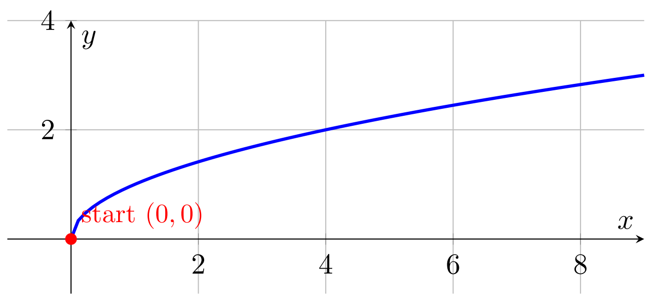
We do together. Name the parent of $f(x) = \sqrt{x}$ and give its domain.

Step 1. Match the rule to a known family: this is the square root parent.

Step 2. Recall its shape: it starts at the point $(0, 0)$ and curves up to the right.

Step 3. A square root needs a non-negative inside, so the domain is $x \geq 0$.

See it:



$y = \sqrt{x}$ — starts at the origin, domain $x \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Name the parent of $f(x) = \dfrac{1}{x}$ and give its domain. reciprocal (rational); domain $x \neq 0$

b) Name the parent of $f(x) = |x|$ and give its range. absolute value; range $y \geq 0$

Summary (fill in): To name a parent: match the rule to a family, recall its shape, then state the domain.

Transformations of Functions **TEACHER KEY**

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Translation: a slide of the graph — $f(x) + k$ moves it up/down, and $f(x - h)$ moves it left/right.

Reflection: a flip: $-f(x)$ flips over the x-axis, and $f(-x)$ flips over the y-axis.

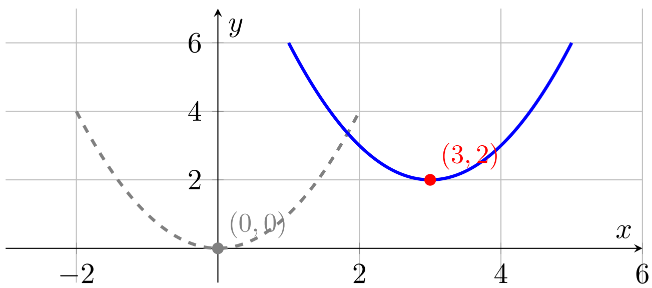
We do together. Describe how $g(x) = (x - 3)^2 + 2$ transforms the parent x^2.

Step 1. Inside the square, $x - 3$ shifts the graph right 3 (opposite the sign).

Step 2. The $+2$ outside shifts it up 2.

Step 3. So the vertex moves from $(0, 0)$ to $(3, 2)$.

See it:



parent (dashed) → right 3, up 2 (solid).

You Try. Solve each. Show every step, then name the answer.

a) Describe $h(x) = -x^2$ from the parent. reflection over the x-axis (opens down)

b) Give the vertex of $y = (x + 1)^2 - 4$. $(-1, -4)$

Summary (fill in): $f(x - h) + k$ shifts a graph h right, k up; a leading $-f(x)$ reflects it over the x-axis.

Domain & Range **TEACHER KEY**

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Domain: all the input values (x) a function is allowed to use.

Range: all the output values (y) the function actually produces.

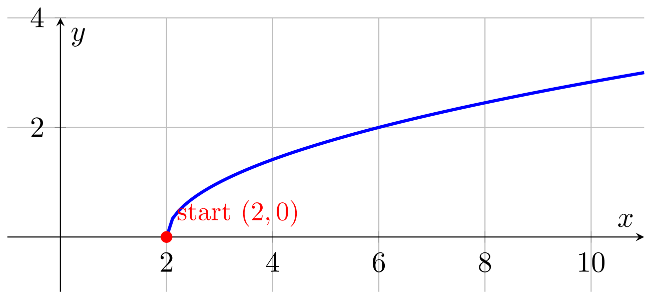
We do together. Find the domain and range of $f(x) = \sqrt{x-2}$.

Step 1. A square root can't hold a negative: set $x - 2 \geq 0$, so the domain is $x \geq 2$.

Step 2. The smallest output is at $x = 2$, where $f(2) = 0$, then it grows.

Step 3. So the range is $y \geq 0$.

See it:



$y = \sqrt{x-2}$ — domain $x \geq 2$, range $y \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Give the domain of $f(x) = \dfrac{1}{x-5}$. $x \neq 5$ (all reals except 5)

b) Give the range of $f(x) = x^2 + 1$. $y \geq 1$

Summary (fill in): Watch two traps: a denominator can't be 0, and a square root can't hold a negative.

Maximum & Minimum **TEACHER KEY**

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Max / min: the turning point of a parabola (the vertex); opens up gives a minimum, opens down gives a maximum.

Increasing / decreasing: read left to right — the graph goes up (increasing) or down (decreasing).

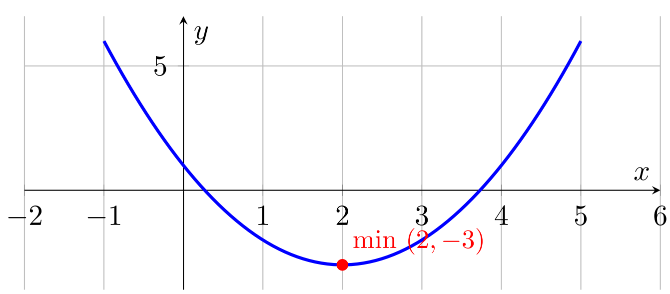
We do together. For $y = (x-2)^2 - 3$, give the vertex, state max or min, and the intervals.

Step 1. Vertex form $a(x-h)^2 + k$ gives vertex $(2, -3)$.

Step 2. Leading coefficient is $+1$ (opens up), so the vertex is a minimum.

Step 3. It decreases for $x < 2$ and increases for $x > 2$.

See it:



opens up $\Rightarrow$ minimum at $(2, -3)$.

You Try. Solve each. Show every step, then name the answer.

a) Give the vertex and max/min of $y = -x^2 + 4$. vertex $(0, 4)$, maximum

b) Give the intervals of $y = x^2$. decreasing $x < 0$, increasing $x > 0$

Summary (fill in): Opens up $\Rightarrow$ minimum; opens down $\Rightarrow$ maximum; the graph turns at the vertex.

Symmetry: Even & Odd Functions

TEACHER KEY

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Even function: symmetric across the y-axis; algebraically $f(-x) = f(x)$.

Odd function: symmetric about the origin; algebraically $f(-x) = -f(x)$.

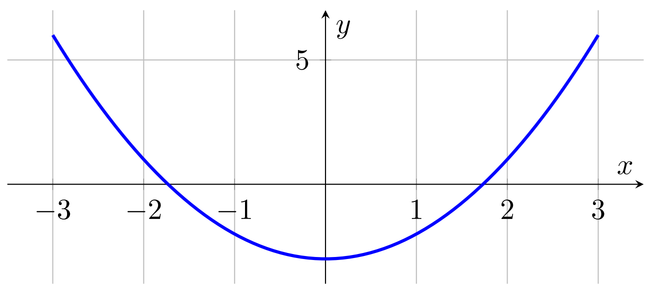
We do together. Is $f(x) = x^2 - 3$ even, odd, or neither?

Step 1. Replace x with $-x$: $f(-x) = (-x)^2 - 3 = \{\{x^2 - 3$—2cm$\}\}$.

Step 2. Compare to $f(x)$: they are equal, so $f(-x) = f(x)$.

Step 3. Therefore f is even (symmetric across the y-axis).

See it (even = y-axis mirror):



$y = x^2 - 3$ — the y-axis is a mirror line (even).

You Try. Solve each. Show every step, then name the answer.

a) Is $f(x) = x^3$ even, odd, or neither? odd: $f(-x) = -x^3 = -f(x)$

b) Is $f(x) = x^2 + x$ even, odd, or neither? neither

Summary (fill in): Compute $f(-x)$: equals $f(x) \Rightarrow$ even; equals $-f(x) \Rightarrow$ odd; otherwise neither.